

Postmortem: Webhook Delivery Backlog in us-iad

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Summary

Beginning at 14:20 UTC on 3 February 2026, event delivery in the us-iad region fell progressively behind. At peak, delivery latency reached 11 hours and 12 minutes. The backlog fully drained at 05:31 UTC on 4 February. No events were lost. The regions eu-tll, eu-fra, and ap-sin were unaffected throughout, and no cross-region failover was triggered because the region remained available and was delivering, merely slowly.

Impact

All 1,204 tenants in us-iad experienced elevated delivery latency to some degree. 88 tenants exceeded four hours of delay. Kirren Retail, whose traffic triggered the condition, was itself among the worst affected, which is a direct consequence of the shared queue architecture in place at the time. Seventeen support tickets were opened during the window. Three customers on the Scale plan received service credits, granted at the discretion of the account owner rather than under a contractual clause, since standard plans carry no credit entitlement.

Timeline

Time (UTC)	Event
14:20	Kirren Retail deploys a change to their receiving endpoint. Endpoint begins returning HTTP 500 for approximately 96 percent of requests.
14:31	Per-region delivery latency probe fires. Primary on-call paged. S2 declared.
15:10	Latency continues climbing. Severity raised to S1.
16:05	Single-tenant origin identified by manually correlating retry volume against tenant IDs in the delivery logs.
16:20	Manual rate limit applied to the Kirren Retail endpoint. Backlog stops growing.
16:45	Kirren Retail engineering contacted directly by the account owner.
18:40	Kirren Retail roll back their endpoint change. Endpoint returns to normal.
18:55	Manual rate limit removed. Backlog begins draining.
05:31 (+1d)	Backlog fully drained. Incident closed.

Table 1. Incident timeline.

Cause

Kirren Retail deployed a change that caused their receiving endpoint to return HTTP 500 for approximately 96 percent of requests while continuing to return 200 for the remaining 4 percent. Relay's retry policy treated these failures as transient, which is correct, and retried each failed delivery up to six times. Because the endpoint was intermittently succeeding, the circuit breaker never opened: at the time it required 20 consecutive failures, and with a 4 percent success rate the longest observed failure run was 14. Retried deliveries were enqueued into the shared regional delivery queue, where they competed for throughput with first-attempt deliveries from every other tenant in the region.

The compounding factor is worth stating explicitly, because it is the part that made an endpoint failure into a regional incident. Each retry that failed generated a further retry, and the retry schedule front-loads attempts within the first thirty seconds. A single tenant with sufficient volume therefore generates, within a minute of failing, several times its normal request rate against a queue sized for normal request rates. The architecture had

no mechanism to prevent one tenant from consuming the shared capacity, and this was a known design limitation rather than a bug; it had been raised in a design review in 2024 and deprioritised on the reasoning that no single tenant was large enough for it to matter.

Detection

The first page fired at 14:31 UTC, eleven minutes after onset, from the per-region delivery latency probe added as action 4 of the Atlas postmortem. Detection worked exactly as intended and no detection improvements are proposed. Diagnosis, however, took a further 94 minutes. The delivery latency dashboard aggregated across all tenants in a region and offered no per-tenant breakdown, so identifying the origin required manually correlating retry volume against tenant IDs in raw delivery logs, a process that one engineer performed by hand while a second continued mitigation.

Resolution

A manual rate limit was applied to the Kirren Retail endpoint at 16:20 UTC, which stopped the backlog growing but did not reduce it. Kirren Retail rolled back their endpoint change at 18:40 UTC after direct contact from their account owner. The remaining backlog drained over the following ten hours without further intervention. Drain throughput was lower than the Atlas incident because the queue was competing with normal weekday traffic rather than draining into an idle overnight period.

Actions

#	Action	Owner	Status
1	Partition delivery queues per tenant with a fair-share scheduler	Platform	Shipped 2026-02-19
2	Change circuit breaker from 20 consecutive failures to a 50 percent failure rate over a rolling 60 second window	Platform	Shipped 2026-02-19
3	Add per-tenant delivery latency to the primary regional dashboard	Platform	Complete 2026-02-11
4	Apply an automatic tenant rate limit when retry volume exceeds 10x that tenant's first-attempt volume	Platform	Shipped 2026-02-19
5	Notify the account owner automatically when a tenant rate limit engages	Platform	Complete 2026-02-24

Table 2. Follow-up actions.

One further observation is recorded here because it did not fit anywhere else. The design limitation that caused this incident was identified nineteen months before it occurred and was deprioritised on an assumption about customer size that had ceased to be true roughly a year later. Nothing in the process re-examined that assumption. The engineering team has since added a standing quarterly review of deprioritised design-review findings, which is not a follow-up action for this incident so much as a response to how the incident became possible.